Computational Études

A Spectral Approach

A Tentative Teaching Plan

Course Information

Course: MATH 794 – 01 (CRN 21296)
Semester: Spring 2026
Schedule: Mon & Wed, 16:30 – 17:45
Location: G-G02013

Instructor

Name: Dr. Denys Dutykh
Department: Mathematics
Institution: Khalifa University
Location: Abu Dhabi, UAE

Legend: Completed Current Examination Planned

Week	Dates	Monday Lecture	Wednesday Lecture
1	Jan 12–14	Course introduction Syllabus & overview	**Ch. 2:** Classical PDEs Heat equation – separation of variables
2	Jan 19–21	**Ch. 3:** Mise en bouche Method of weighted residuals, collocation example, collocation vs Galerkin – Project I assigned	**Ch. 4:** The Geometry of Nodes Lagrange interpolation, Runge phenomenon, potential theory
3	Jan 26–28	**Ch. 5:** Differentiation Matrices FD stencils, spectral matrices, Fornberg algorithm	**Ch. 6:** Smoothness and Spectral Accuracy Smoothness, spectral convergence, exponential accuracy
4	Feb 2–4	**Ch. 7:** Chebyshev Differentiation Matrices Chebyshev nodes, D_N matrix, negative sum trick	**Ch. 8:** Boundary Value Problems 1D/2D BVPs, matrix stripping, Newton iteration – Project II assigned
5	Feb 9–11	**Project I:** Students Presentations Oral presentations and discussions of Project I results	**Ch. 9:** Fourier Space on Grids Fourier transform, DFT/FFT, aliasing, sinc interpolation

Week	Dates	Monday Lecture	Wednesday Lecture
6	Feb 16–18	**Ch. 10:** Spectral PDE Solvers Chebyshev–Fourier recap, Chebyshev differentiation via FFT, method of lines	**Ch. 11:** Pseudo-spectral PDE Solvers Fourier Pseudospectral Methods – Project III assigned
7	Feb 23–25	**Ch. 12** TBA	**Ch. 13** TBA
8	Mar 2–4	**Project II:** Students Presentations Oral presentations and discussions of Project II results	**Midterm Examination**
9	Mar 9–11	**Ch. 14** TBA	**Project III:** Students Presentations Oral presentations and discussions of Project III results
–	Mar 16–27	*Spring Break & Eid Al Fitr – No classes*	
10	Mar 30–Apr 1	TBA	TBA
11	Apr 6–8	TBA	TBA
12	Apr 13–15	TBA	TBA
13	Apr 20–22	TBA	TBA
14	Apr 27–29	TBA	TBA
–	May 4–14	**Final Examinations Period**	

Table 1: Teaching schedule for MATH 794 – Spring 2026

Notes

- This schedule is tentative and may be adjusted based on class progress.
- Chapter numbers refer to *Computational Études: A Spectral Approach*.
- Office hours are available by appointment.
- All course materials are available in the course repository.

Course Projects

Project I: Spectral Methods for Boundary Value Problems (Based on Chapter 3 – Assigned: Jan 19, 2026)

Implement spectral methods (collocation and/or Galerkin) to solve a boundary value problem of your choice. Requirements:

1. Find a linear BVP of at least second order with **non-homogeneous** boundary conditions.
2. Construct an exact solution to this problem (or design the problem around a known exact solution).
3. Choose appropriate basis functions that satisfy the boundary conditions identically.
4. Apply the **Collocation** method, **Galerkin** method, or both to obtain a numerical solution.
   - If using **only** the Collocation method, the BVP must have **non-constant coefficients**.
   - If applying **both** methods, the problem may have constant coefficients (though non-constant coefficients are also welcome).
5. Present results in a table that includes the error at the collocation points.
6. Provide graphical representations showing: the exact solution, the numerical approximation, and the difference (error) between them.

Deliverables: A written report, a short oral presentation (15 min max), and a live demonstration of your computational codes. Grading is based on relative performance within the class.

Project II: Spectral Collocation for BVPs (Based on Chapters 6–8 – Assigned: Feb 2, 2026)

Implement Chebyshev spectral collocation to solve boundary value problems. Requirements:

1. Implement the Chebyshev differentiation matrix D_N using the negative sum trick.
2. Choose a second-order BVP (linear or nonlinear) with known exact solution.
3. Solve the BVP using spectral collocation with matrix stripping for boundary conditions.
4. Present a convergence study: error vs. N on a semilog plot.
5. For bonus: solve a 2D problem using tensor products.

Deliverables: A written report, a short oral presentation (15 min max), and a live demonstration of your computational codes. Grading is based on relative performance within the class.

Project III: Fourier Pseudospectral Simulation of Periodic PDEs (Based on Chapters 9–11 – Assigned: Feb 18, 2026)

Implement a Fourier pseudospectral method to simulate the dynamics of a PDE of your choice on a periodic domain. Requirements:

1. Choose a time-dependent PDE in (x, t) variables posed on a periodic spatial domain.
   - The PDE may be linear (e.g. advection, diffusion, Schrödinger) or nonlinear (e.g. Korteweg–de Vries, Burgers, Allen–Cahn, Kuramoto–Sivashinsky).
2. Discretize spatial derivatives using the Fourier pseudospectral method (via the FFT).
3. Advance in time using an appropriate ODE solver (e.g. Runge–Kutta, exponential integrator, or split-step method).

4. Validate your implementation against an exact solution, a conserved quantity, or a known qualitative behaviour of the chosen PDE.
5. Present space-time visualizations of the computed dynamics (e.g. waterfall plots, animations, or snapshots at selected times).
6. For bonus: extend the simulation to two or three spatial dimensions.

Deliverables: A written report, a short oral presentation (15 min max), and a live demonstration of your computational codes. Grading is based on relative performance within the class.

Project IV

To be announced

Last updated: February 16, 2026